Military Specification

Propellant, Liquid Neon

This specification is approved for use by the Air Force Rocket Propulsion Laboratory (LKCP), Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the requirements for liquid neon.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-27626	Sampler, Liquid Oxygen, TTU-131/E

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-172	Color Code for Containers of Liquid Propellants

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

DEPARTMENT OF TRANSPORTATION (DOT)

49 CFR 170-179	Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(Applications for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20402.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to the Air Force Rocket Propulsion Laboratory, LKCP, Edwards AFB CA 93523, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9135

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[redacted] SOCIETY FOR TESTING AND MATERIALS (ASTM)

F310-70	Sampling Cryogenic Aerospace Fluids
E29-73	Recommended Practices for Indicating which Figures are to be Considered Significant in Specified Limiting Values

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

NATIONAL BUREAU OF STANDARDS (NBS)

Research Paper	Measurement of Water in Gases by Electrical Conduction in a Film of Hygroscopic Material and the Use of Pressure Changes in Calibration

3. REQUIREMENTS

3.1 Composition. The composition of the neon shall conform to the limits in Table I when tested in accordance with the applicable test methods (4.5).

Table I, Requirements

Component	Limit	Test Method
Assay (%/vol)	99.95 Min	4.5.8
Nitrogen (ppm/vol)	5 Max	4.5.1
Oxygen (ppm/vol)	1 Max	4.5.2
Helium (ppm/vol)	500 Max	4.5.3
Hydrogen (ppm/vol)	4 Max	4.5.4
Carbon dioxide plus Carbon Monoxide (ppm/vol)	0.5 Max	4.5.5
Total Hydrocarbons (ppm/vol)	1 Max	4.5.6
Water (ppm/vol)	1 Max	4.5.7

3.2 Limiting Values. The following applies to all specified limits in this specification: for purposes of determining conformance with these requirements, an observed value or a calculated value shall be rounded off "to the nearest unit" in the last right hand place of figures used in expressing the limitation value, in accordance with ASTM Method E29-73, "Recommended Practices for Indicating Which Figures are to be Considered Significant in Specified Limiting Values."

3.3 Filters. The liquid neon product shall be filtered through a 10 micron nominal (25 micron absolute) filter located in the fill line to the shipping containers.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements (6.2).

4.1.1 Point of Inspection. Unless otherwise specified, quality conformance tests of the liquid neon shall be conducted at the site of filling. The quality conformance tests shall be completed prior to shipment of filled containers.

4.2 Classification of Tests. The inspection and testing of the propellant shall be classified as quality conformance tests.

4.3 Quality Conformance Tests. Quality conformance tests are defined as tests specified herein to assure conformance to the requirements of Table I of this specification.

4.3.1 Lot. All containers filled at the same time from a single source shall constitute a lot.

4.3.2 Sampling. A number of Neon containers shall be selected in accordance with Table II and sampled. The resulting samples shall be subjected to tests described in 4.5.

TABLE II, SAMPLING FOR TEST

NUMBER OF CONTAINERS IN LOT	NUMBER OF CONTAINERS TO BE SAMPLED
1	1
2-40	2
41-70	3
71-over	4

4.3.2.1 Sample. Each sample shall be of sufficient size to conduct all the quality conformance tests as specified herein. Unless otherwise specified, the quality conformance tests shall be made of each required sample (6.2). When required, an equivalent sample shall be forwarded to a laboratory designated by the procuring activity for subjection to the quality conformance tests specified herein

4.3.2.2 Samplers The sampler for liquid neon shall be in accordance with MIL-S-27626 or functionally equivalent thereto. The liquid samplers convert the entrapped liquid to gas. The aliquots taken for analysis are representative samples.

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4.3.2.3 Sampling Methods. Unless otherwise specified, the neon shall be sampled in accordance with ASTM F310-70 (6.2).

4 3.3 Liquid Cylinders. The number of liquid cylinders (Dewar type) filled with liquid neon selected for sampling for each lot shall be in accordance with Table II. The first and last cylinders to be filled within a given lot shall be sampled. Other samples may be selected at random. The neon from each cylinder sampled shall constitute a separate sample. For the purposes of selecting sample cylinders only, any one cylinder may be selected from a group of cylinders filled simultaneously from a single manifold.

4.4 Rejection. When any sample of the propellant, pressurizing agent tested in accordance with 4.5 fails to conform to the requirements specified herein, the entire lot represented by the sample shall be rejected (6.2).

4.5 Test Methods.

4.5.1 Nitrogen Content. The nitrogen content shall be determined by the following test method.

4.5.1.1 A gas chromatograph equipped with a Molecular Sieve (Type 5Å) column, helium carrier gas, and a detector capable of measuring directly the nitrogen content of the sample at the specified limit shall be used. If a less sensitive detector is used, appropriate pre-concentration of the nitrogen in the sample may be accomplished using an activated coconut charcoal-filled trap maintained at the temperature of liquid nitrogen. The gas chromatograph used shall be capable of separating and determining the nitrogen content of the sample with sensitivity of at least 20% of the specified limit.

4.5.2 Oxygen Content. The oxygen content of the sample shall be determined by a specific oxygen analyzer which operates on an electrolytic, galvanic cell (e.g. polarography), or heat of reaction principle of measurement. Calibration of the analyzer may be performed with accurate trace oxygen standards, or integrally in accordance with Faraday's Law. The sensitivity and accuracy of the instrument, when calibrated, shall not be less than plus or minus five percent of the specified maximum oxygen content, measuring accuracy shall not be less than two times the sensitivity. In cases of dispute, the galvanic cell shall be used as the referee.

4.5.3 Helium Content. The helium content shall be determined using a gas chromatograph equipped with a Molecular Sieve (Type 5Å) column, neon carrier gas (99.99% neon minimum), and a thermal conductivity detector. The gas chromatograph used shall be capable of separating and determining the helium content of the sample with a sensitivity of at least 20% of the specified limit.

4.5.4 Hydrogen Content. The hydrogen content of the sample shall be determined by one of the following methods. In cases of dispute, the heat of reaction method in 4.5.4.3 shall be used as the referee.

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4.5.4.1 A gas chromatograph method wherein hydrogen can be adequately separated and specifically measured at the limit specified in Table I.

4.5.4.2 By an analyzer in which hydrogen reacts to form a compound which is subsequently measured. The analyzer shall be calibrated at appropriate intervals by the use of calibration gas standards (6.4). The range of the analyzer shall be no greater than ten times the specified maximum hydrogen content.

4.5.4.3 By a heat of reaction type analyzer. The analyzer shall be calibrated at appropriate intervals by the use of calibration gas standards or integrally in accordance with Faraday's Law. The range of the analyzer shall be no greater than ten times the specified maximum hydrogen content.

4.5.5 Carbon Dioxide plus Carbon Monoxide Content. The carbon dioxide plus carbon monoxide content shall be determined by using either a non-dispersive IR which can distinguish between CO and CO_2 or a gas chromatograph equipped with an Ethyl vinyl benzene polymer bead column ("Porapak Q", "Chromosorb 102", etc.), and a detector capable of measuring directly each of the above components. If a less sensitive detector is used, appropriate preconcentration of the carbon dioxide and carbon monoxide may be accomplished using a trap filled with ethyl vinyl benzene polymer beads maintained at the temperature of liquid nitrogen. The gas chromatograph used shall be capable of separating and determining the carbon dioxide and carbon monoxide content of the sample with a sensitivity of at least 20% of the specified limit. Referee should be NDIR.

4.5.6 Total Hydrocarbons Content. Cylinder-filled propellant, pressurizing agent samples or samplers for container-filled propellant, pressurizing agent shall be connected to the inlet of a total hydrocarbon analyzer meeting the requirements of 4.5.6.1. Sufficient neon shall be permitted to flow to obtain a stabilized reading.

4.5.6.1 Total Hydrocarbon Analyzer. The total hydrocarbon analyzer shall be operated on some method such that all hydrocarbons in the sample stream shall be reduced to a common reference. The common reference shall be quantitatively analyzed and indicated in ppm by weight as carbon or in ppm by volume as methane equivalent, or in any other indication that can readily be converted to these units. The analyzer shall be sensitive to differences of plus or minus 0.5 ppm at and below the limits specified herein.

4.5.7 Water Content. The sample shall be tested for water content in accordance with one of the following methods. The electrolytic method of 4.5.7.3 shall be the referee method.

4.5.7.1 Electrical Conductivity Method. Use in accordance with NBS Research Paper RP 1865.

4.5.7.2 Dew Point Method

4.5.7.3 Electrolytic Method.

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4.5.8 Neon Content. Assay of the neon shall be computed by subtraction of the aggregate impurity sum from 100 percent.

5. PREPARATION FOR DELIVERY

5.1 Packaging Unless otherwise specified, liquid neon shall be packaged in Dewars conforming to DOT 49 CFR 178.57.

5.2 Preparation of Shipping Containers. The contractor shall establish the condition of all containers and insure that 5.3 is complied with. Contractor owned containers shall be cleaned, tested, and repaired in accordance with the schedule established in the contract or purchase order (6.2). All containers shall be free from oils, greases, and other odorous or toxic impurities.

5.3 Liquid Cylinders. Qualification, maintenance, and use of liquid cylinders (Dewar type) shall conform to requirements of DOT 49 CFR 173.34, any other DOT requirement, and as follows.

5 3 1 Cleaning and Repair. Unless otherwise provided for in the contract or purchase order, all cylinders shall be visually inspected both internally and externally for the presence of water, rust, scale, oil film, or other foreign matter, sufficient insulation and vacuum, and physical damage (6 2). Any insulation damage, vacuum depletion, or physical damage which would endanger safe transportation of the product shall be repaired prior to reuse. If evidence of internal contamination is found, the cylinders shall be recleaned by a suitable method to remove the contamination prior to use.

5.3.2 Gaskets. Gaskets used to seal cylinder openings shall be polytetrafluoroethylene, polychlorotrifluoroethylene or other material compatible with the product and approved for use by the procuring activity (6.2) The contractor shall assure that all gaskets are serviceable and furnish new gaskets when necessary so that a tight seal is assured.

5.3.3 Filling. Cylinders shall be filled through a filter (3 3) to the filling density specified by DOT.

5.4 Labeling and Marking. All government-owned containers shall be marked in accordance with MIL-STD-129. In addition, an identification tag, and container color code shall be used.

5.4.1 Identification Tag. Unless otherwise specified in the contract or purchase order, an identification tag impervious to climate conditions shall be wired to the outlet port of each container and shall contain the following information product name, specification number with revision letter, NSN number, quantity, name of manufacturer, name of contractor (if different from manufacturer), date of manufacture, and lot identification number (6.2)

5.4.2 Cylinder Color Code Unless otherwise specified by the procuring activity, each cylinder shall be color coded in accordance with MIL-STD-172,

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and the exact name identification to be marked on the outside of the cylinders shall be liquid neon (6.2). Any other name identification shall be obliterated by removing and overpainting.

6.0 NOTES This section contains non-mandatory provisions only to assist both the contractor and buyer in the proper understanding and utilization of this specification.

6.1 Intended Use Liquid neon is used as a cryogenic coolant to simulate space environment, in lasers, in radars, and for other cryogenic purposes

6.2 Ordering Data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement Requirements.

6.2.1.1 Title, number, type, and date of this specification.

6.2.1 2 Method of shipment, type and capacity of container (5 1).

6.2.1 3 When inspection requirements are to be performed by other than the supplier (4.1).

6 2.1 4 When waiver on quality conformance tests on each sample is granted (4.3.2.1).

6 2.1.5 When variation in sampling method is granted (4.3 2 3).

6.2.1.6 When disposition of rejected product has to be specified (4.4).

6.2.1 7 When cleaning and repair schedule is required for leased or Government owned containers (5.2)

6.2.1.8 When cleaning and repair provisions are other than specified (5.3.1).

6.2.1 9 When approval of gasket material is required (5 3.2).

6.2.1.10 When identification tag is other than as specified (5 3.1)

6.2.1.11 When the container color code is waived (5.4 2).

6.2.1.12 When variation in requirement is granted (3.1).

6.2.2 Contract Data Requirements Data conforming to Data Item Description DI-T-3733, Quality Conformance Test Reports (Fuels), is a requirement for delivery in connection with this specification. The data item will be specified for delivery on DD 1423.

6.3 Supersession Data. This specification includes the requirements of Air Force Purchase Item Description (AFPID) 9135-16 dated 6 February 1969.

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Air Force - 12

REVIEW ACTIVITY

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PREPARING ACTIVITY

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A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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3 IS THE DOCUMENT RESTRICTIVE?

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